Ahmed Oualid Bahri

AI Engineer | Full-Stack Developer | Python Coach
Mostaganem, Algeria (Open to Remote) | +213 652 12 90 11
bahriahmedwalid@gmail.com
GitHub: github.com/OualidML | LinkedIn: linkedin.com/in/ahmed-oualid-bahri



PROFESSIONAL SUMMARY

National Polytechnic School of Constantine engineering student and BrainerX AI graduate who bridges mathematically rigorous deep learning with production-grade full-stack engineering. Builds calibrated, safety-first multimodal models and ships them as offline-capable React/TypeScript and Streamlit applications for high-stakes domains such as clinical decision support. Mentors **60+** engineering students as Python Coach and is seeking remote AI Engineering and Full-Stack roles.

TECHNICAL SKILLS

- **AI / Machine Learning:** Deep Learning (CNNs, EfficientNet), Multimodal Fusion, Probability Calibration (Temperature Scaling), Gradient Boosting (LightGBM), SHAP Explainability, Cross-Validation, Hyperparameter Optimization, Clustering (K-Means, DBSCAN, Hierarchical), PCA, Model Evaluation
- **Languages:** Python, TypeScript, JavaScript
- **Frontend / Full-Stack:** React, Tailwind CSS, Progressive Web Apps (PWA), Streamlit
- **Data / Storage:** Dexie.js, IndexedDB, Offline-First Architecture, B-Tree Indexing, Data Analysis Pipelines
- **AI Integration:** Google Gemini API, Generative AI Tools (VEO-3, Higgsfield)
- **Foundations:** Advanced Mathematics, Physics, Algorithms and Data Structures, Code Optimization
- **Professional:** Technical Communication, Mentoring, Problem Solving

HIGH-IMPACT PROJECTS

ORDO-DZ Clinical Prescription Engine | Full-Stack / PWA

React, TypeScript, Tailwind CSS, Dexie.js, IndexedDB

- Architected an offline-first clinical prescription PWA, as measured by a local client-side database indexing **16,700+** standardized pharmaceutical catalog entries with **100%** offline reliability, by building on React, TypeScript, Tailwind CSS, Dexie.js, and IndexedDB.
- Engineered a geocoded registry of **5,300+** licensed pharmacies spanning **58** Algerian wilayas, as measured by instant in-app search with no server dependency, by embedding the dataset directly in the client-side database.
- Reduced drug and pharmacy lookup times to **<15ms** with **100%** offline reliability and **0 DZD** server maintenance overhead, as measured by in-browser query benchmarks, by applying Dexie.js and IndexedDB B-Tree indexing.
- Delivered standardized medical document generation (Chifa, Sécurisée) for doctors, as measured by fully offline operation, by implementing a local-first PWA architecture.

Multimodal Skin Lesion Diagnostic Assistant | Deep Learning

EfficientNetB0, Temperature Scaling, Google Gemini API, Streamlit

- Engineered a multimodal joint fusion architecture, as measured by **1,280**-dimensional visual embeddings from EfficientNetB0 fused with **3** clinical metadata features across **10,015** dermoscopic images, by combining image and tabular patient data in a single classifier.
- Calibrated neural overconfidence, as measured by predicted probabilities aligned with empirical probabilities, by implementing post-hoc Temperature Scaling (**T = 1.5909**).
- Formulated an asymmetric clinical safety cutoff under a zero-false-negative mandate, as measured by capturing **96%** of malignant melanomas, by setting a **15.0%** malignant probability safety boundary.
- Deployed an explainable diagnostic application, as measured by automated clinical rationales for flagged cases, by integrating the Google Gemini API into a Streamlit interface.

Vehicle Price Prediction Engine | Machine Learning

LightGBM, SHAP, Cross-Validation

- Engineered an end-to-end regression pipeline, as measured by **86%** overall model accuracy on a dataset of **1,000,000+** vehicle records across **25** multimodal tabular features, by building and tuning a LightGBM model.
- Benchmarked model reliability and transparency, as measured by **5**-fold cross-validation, hyperparameter optimization, and SHAP explainability, by applying a rigorous validation workflow.

SuperManager Pro | Offline Enterprise POS Engine
Desktop / PWA, Generative AI Media

- Architected a **100%** offline desktop/PWA point-of-sale system, as measured by **zero** cloud dependency for retail environments, by designing a local-first data layer for inventory control and sales analytics.
- Optimized checkout speed, as measured by instantaneous barcode-to-cart transaction processing with **zero** latency, by implementing digital barcode scanning against local storage.
- Produced multi-scene AI marketing assets, as measured by complete promotional video ads, by directing generative AI tools (VEO-3, Higgsfield).

UCI Adult Income ML Capstone | Data Science
K-Means, DBSCAN, Ward Hierarchical Clustering, PCA

- Benchmarked clustering architectures (K-Means, DBSCAN, Ward Hierarchical) and PCA dimensionality reduction, as measured by **82%** overall income-prediction accuracy across **48,842** records, by isolating the critical demographic predictors of income.

ENGINEERING EXPERIENCE & LEADERSHIP

Python Coach | Cirtec Engineering Club, National Polytechnic School of Constantine | Apr 2026 - Jun 2026

- Mentored **60+** polytechnic engineering students, as measured by **40+** hours of hands-on workshops, by delivering structured training in Python algorithmic optimization, data structures, and production-grade ML workflows.
- Accelerated student progress from working code to efficient code, as measured by improved algorithmic performance in workshop exercises, by coaching code optimization and complexity analysis.
- Translated complex technical concepts into practical explanations for mixed-experience groups, as measured by sustained participation across **40+** hours of sessions, by tailoring curriculum depth to each audience.

EDUCATION

- **Second Preparatory Year, Engineering** | National Polytechnic School of Constantine (ENP Constantine) | Sep 2025 - Present
- Coursework: Advanced Mathematics, Physics, Electronics, Mechanical Engineering, Computer Science
- **Baccalaureate, Mathematics (High Distinction)** | Lycée, Mostaganem, Algeria | Jun 2025

CERTIFICATIONS

- **AI Masterclass Certification** | BrainerX Tech Academy | October 2026
- **Python Coach, Cirtec Engineering Club** | ENP Constantine | April 2026 - June 2026 (2 months)
- **TCF DAP, French Academic Proficiency Test (B2)** | Official French Language Certification | Nov 2025

LANGUAGES

- Arabic: Native
- English: Fluent (Advanced)
- French: B2 (TCF DAP, Nov 2025)